Integrated motor drivers, dual buck supplies, encoder conditioning and onboard RP2040.
EV3 nominal hole geometry from user drawing; pinmap/current HOLD. Manufacturing release is blocked.
Use electronics/README.md and generated/electronics/release-status.json.

controller

File: controller.kicad_sch

controller_power

File: controller_power.kicad_sch

usb

File: usb.kicad_sch

flash_clock

File: flash_clock.kicad_sch

power

File: power.kicad_sch

buck1

File: buck1.kicad_sch

buck2

File: buck2.kicad_sch

arm_R

File: arm_R.kicad_sch

arm_L

File: arm_L.kicad_sch

arm_F

File: arm_F.kicad_sch

arm_B

File: arm_B.kicad_sch

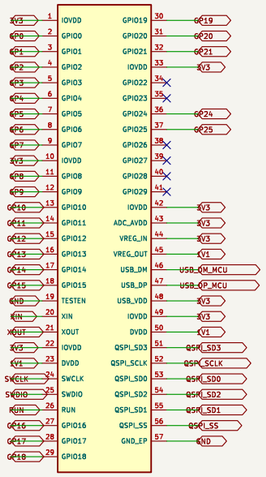
gonewx/quadcuber

Sheet: /
File: quadcuber.kicad_sch

Title: Quadcuber four-arm integrated controller - A1 ENGINEERING HOLD

Size: A3	Date:	Rev: A1
KiCad E.D.A. 9.0.2		Id: 1/12

U1



RP2040

J2



UART GND TX RX

J3



RESET

J4



SWD VREF D G C

R400



10k

S2



RUN RESET

C400



100n

R405



100k

R406



100k

D7



USER LED GP25

R410



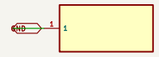
1k

F2



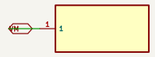
0.5A encoder PTC

TP1



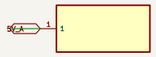
GND

TP2



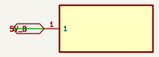
VM

TP3



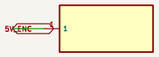
5V_A

TP4



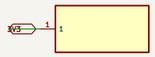
5V_B

TP5



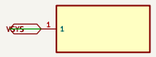
5V_ENC

TP6



3V3

TP7



VSYS

Sheet: /controller/
File: controller.kicad_sch

Title: Quadcuber A1 / controller / ENGINEERING HOLD

Size: A3	Date:	**Rev: A1**
KiCad E.D.A. 9.0.2		Id: 2/12















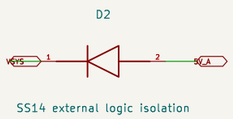


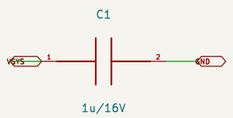


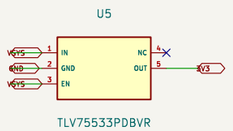


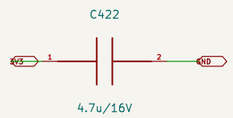












Sheet: /controller_power/
File: controller_power.kicad_sch

Title: Quadcuber A1 / controller_power / ENGINEERING HOLD

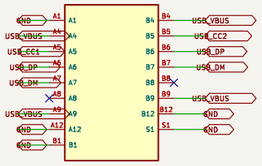
Size: A3	Date:	**Rev: A1**
KiCad E.D.A. 9.0.2		Id: 3/12

D5



SS14 USB isolation

J5



USB-C data / BOOTSEL

R401



5.1k

R402



5.1k

F3



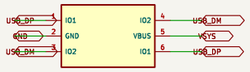
0.5A USB PTC

C423



1u/16V

U6



USBLC6-2SC6

R403



27

R404



27

C427

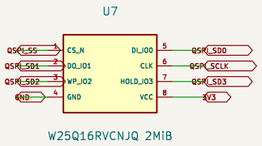


100n

Sheet: /usb/
File: usb.kicad_sch

Title: Quadcuber A1 / usb / ENGINEERING HOLD

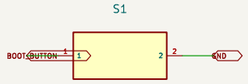
Size: A3	Date:	**Rev: A1**
KiCad E.D.A. 9.0.2		Id: 4/12

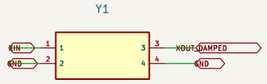






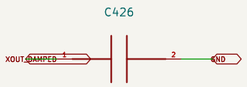












Sheet: /flash_clock/
File: flash_clock.kicad_sch

Title: Quadcuber A1 / flash_clock / ENGINEERING HOLD

Size: A3	Date:	**Rev: A1**
KiCad E.D.A. 9.0.2		Id: 5/12

J1



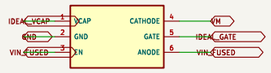
9V regulated input

F1



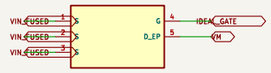
8A 24V time-lag fuse

U4



LM74700QDBVRQ1

Q1



CSD18540Q5B

C4



100n

C5



100n

D4

SMBJ12CA input transient

D3

SMBJ12A transient clamp

C2

470u/25V

C3

470u/25V

R500

300R 1W VM bleed

R501

300R 1W VM bleed

#FLG1

PWR_FLAG

#FLG2

PWR_FLAG

#FLG3

PWR_FLAG

#FLG4

PWR_FLAG

#FLG5

PWR_FLAG

#FLG6

PWR_FLAG

#FLG7



PWR_FLAG

#FLG8



PWR_FLAG

#FLG9

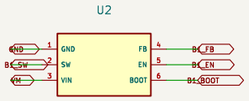


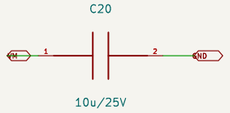
PWR_FLAG

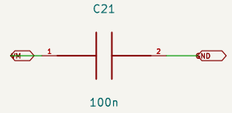
Sheet: /power/
File: power.kicad_sch

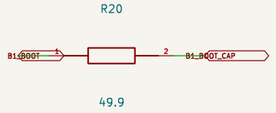
Title: Quadcuber A1 / power / ENGINEERING HOLD

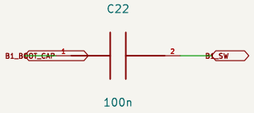
Size: A3	Date:	**Rev: A1**
KiCad E.D.A. 9.0.2		Id: 6/12

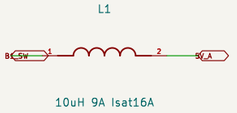


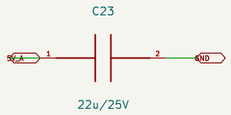


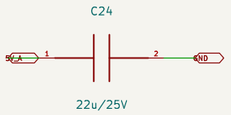


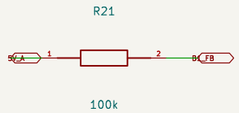


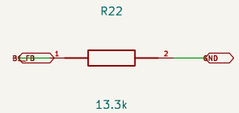


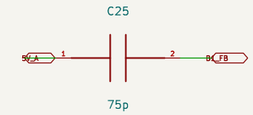


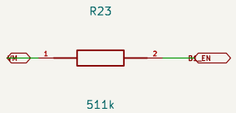


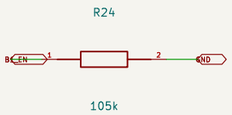


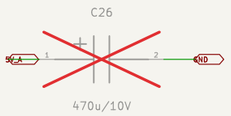








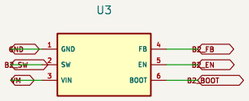


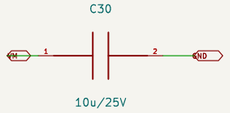


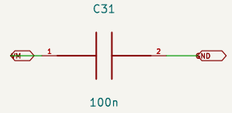
Sheet: /buck1/
File: buck1.kicad_sch

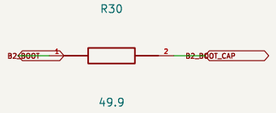
Title: Quadcuber A1 / buck1 / ENGINEERING HOLD

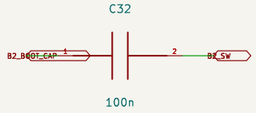
Size: A3	Date:	**Rev: A1**
KiCad E.D.A. 9.0.2		Id: 7/12

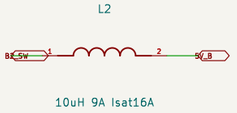


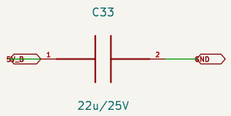


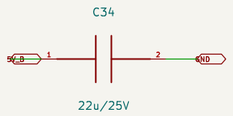


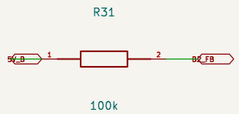


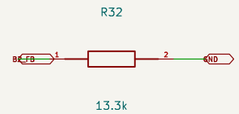


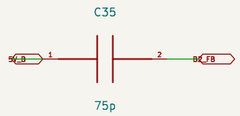


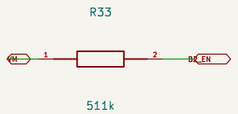


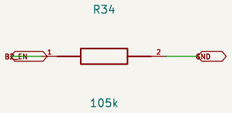


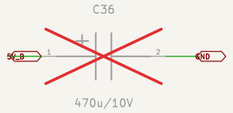










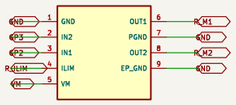


Sheet: /buck2/
File: buck2.kicad_sch

Title: Quadcuber A1 / buck2 / ENGINEERING HOLD

Size: A3	Date:	**Rev: A1**
KiCad E.D.A. 9.0.2		Id: 8/12

U10



DRV8871DDAR

R100



43.2k

R101



100k

R102



100k

C100



100n

C101



10u/25V

C102



470u/25V

J10



EV3 R PINMAP HOLD

U20



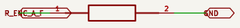
SN74LVC2G17DBVR

R103



1k

R105



100k

R104



1k

R106



100k

C103



100n

C104



100n

F10



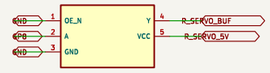
1.1A servo PTC

J20



R SERVO G + S

U30



SN74AHCT1G125DBVR

C105



100n

R109



100k

R107



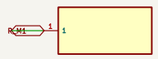
220

R108



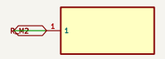
100k

TP10



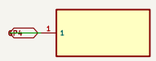
R_M1

TP11



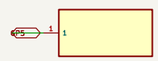
R_M2

TP12



GP4

TP13

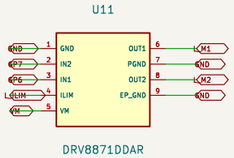


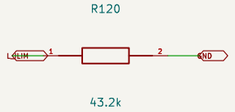
GP5

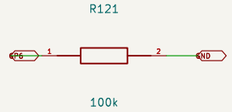
Sheet: /arm_R/
File: arm_R.kicad_sch

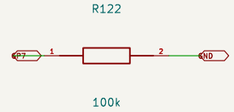
Title: Quadcuber A1 / arm_R / ENGINEERING HOLD

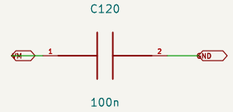
Size: A3	Date:	**Rev: A1**
KiCad E.D.A. 9.0.2		Id: 9/12

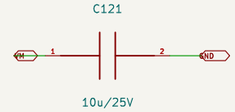


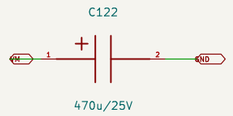


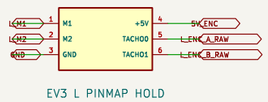




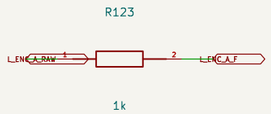


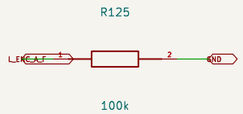


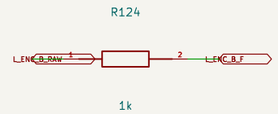


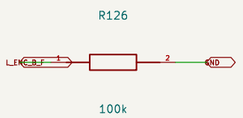


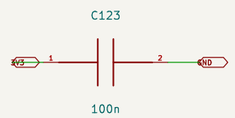


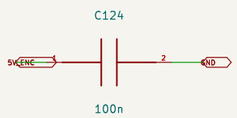


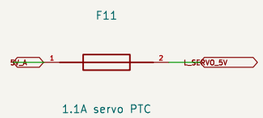


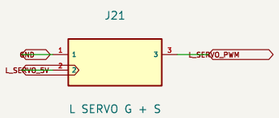


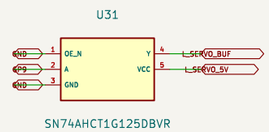


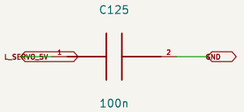


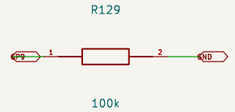


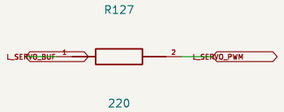


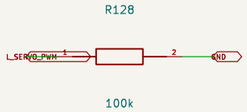


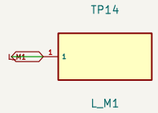


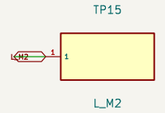


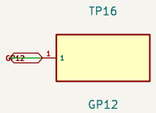


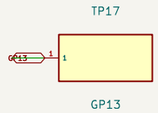










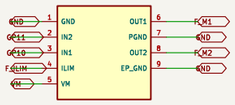


Sheet: /arm_L/
File: arm_L.kicad_sch

Title: Quadcuber A1 / arm_L / ENGINEERING HOLD

Size: A3	Date:	**Rev: A1**
KiCad E.D.A. 9.0.2		Id: 10/12

U12



DRV8871DDAR

R140



43.2k

R141



100k

R142



100k

C140



100n

C141



10u/25V

C142

470u/25V

J12

EV3 F PINMAP HOLD

U22

SN74LVC2G17DBVR

R143

1k

R145

100k

R144

1k

R146

100k

C143

100n

C144

100n

F12

1.1A servo PTC

J22

F SERVO G + S

U32

SN74AHCT1G125DBVR

C145

100n

R149

100k

R147

220

R148

100k

TP18

F_M1

TP19

F_M2

TP20

GP16

TP21

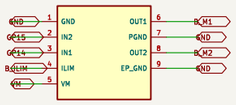
GP17

Sheet: /arm_F/
File: arm_F.kicad_sch

Title: Quadcuber A1 / arm_F / ENGINEERING HOLD

Size: A3	Date:	**Rev: A1**
KiCad E.D.A. 9.0.2		Id: 11/12

U13



DRV8871DDAR

R160



43.2k

R161



100k

R162



100k

C160



100n

C161



10u/25V

C162



470u/25V

J13



EV3 B PINMAP HOLD

U23



SN74LVC2G17DBVR

R163



1k

R165



100k

R164



1k

R166



100k

C163



100n

C164



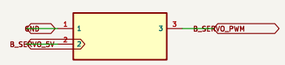
100n

F13



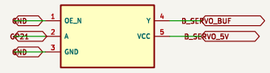
1.1A servo PTC

J23



B SERVO G + S

U33



SN74AHCT1G125DBVR

C165



100n

R169



100k

R167



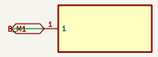
220

R168



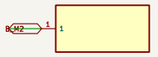
100k

TP22



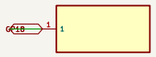
B_M1

TP23



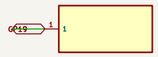
B_M2

TP24



GP18

TP25



GP19

Sheet: /arm_B/
File: arm_B.kicad_sch

Title: Quadcuber A1 / arm_B / ENGINEERING HOLD

Size: A3	Date:	**Rev: A1**
KiCad E.D.A. 9.0.2		Id: 12/12